

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Data Structures and Algorithms		
Core-I – Theory	I Year & I Semester	Credit:4	436C1A

Objectives:

- Define the basic concepts of algorithms and analyze the performance of algorithms.
- Discuss various algorithm design techniques for developing algorithms.
- Discuss various searching, sorting and graph traversal algorithms.
- Understand NP completeness and identify different NP complete problems.
- Discuss various advanced topics on algorithms.

Outcomes:

1.	Analyze programming problem statements.	K1, K2
2.	Comprehend and select algorithm design approaches in a problem specific manner.	K2, K3
3.	Choose appropriate data structures for a specific problem	K3, K4
4.	Utilize necessary mathematical abstractions to solve problems	K5, K6
5.	Come up with analysis of efficiency and proofs of correctness	K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I: Introduction: Algorithm, Pseudo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, Asymptotic Notation- Big oh notation, Omega notation, Theta notation and Little oh notation, Probabilistic analysis, Amortized analysis.

UNIT II: Insertion and deletion and merging with 1) binary search tree, 2) AVL tree, 3) Red Black tree, 4) B tree, 5) B+ tree and Comparison of previous tree structures. Fibonacci Heap, Fibonacci Heap Operations: Find minimum, merge, insert, extract minimum, decrease key and delete, Complexity analysis of the above data structure operations.

UNIT III: Representations of Graphs, Minimum Spanning Trees: Growing a Minimum Spanning Tree – Kruskal and Prim- Single-Source Shortest Paths: The Bellman-Ford algorithm – Single-Source Shortest paths in Directed Acyclic Graphs – Dijkstra ‘s Algorithm, Divide and conquer: General method, applications - Quick sort, Merge sort, Strassen’s matrix multiplication, External Sort: External merge sort, K-Way Merge sorting

UNIT IV: Greedy method: General method, applications-Job sequencing with deadlines, 0/1, knapsack problem, Huffman Codes, Dynamic Programming: General method, applications-Matrix chain multiplication, 0/1 knapsack problem, Traveling salesperson problem, Reliability design.

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

UNIT V: Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Branch and Bound: General method, applications - Traveling salesperson problem, 0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution. NP-Hard and NP-Complete problems

Recommended Texts:

1. Peter Brass; Advanced Data Structures; CAMBRIDGE UNIVERSITY PRESS;2008
2. S. Dasgupta, C. Papadimitrou, U Vazirani; Algorithms; Mc Graw Hill;2022
3. J. Klienbergr and E. Tardos, Algorithm Design, Pearson Education Limited;2013.
4. Ellis Horowitz, Sartaj Sahni, Rajasekharan, Fundamentals of Algorithms, 2nd Edition, Universities Press, 2009.

Reference Books:

1. Sartaj Sahni, Data Structures Algorithms and Applications in C++, 2nd Edition, Universities Press, 2007.
2. Aho V Alfred, Hapcroft E John, Ullman D Jeffry, Data Structures and Algorithms, Pearson Education, 2001.
4. Adam Drozdek, Thomson, Data Structures and Algorithms in JAVA, 3rd Edition, Cengage Learning, 2008.
5. Horowitz, Sahni, Mehta, Fundamentals of Data Structures in C++, 2nd Edition, Universities Press, 2007.

Web References:

1. <https://nptel.ac.in/courses/106102064>

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	M	S	S	S	L	M
CO 2	S	L	S	M	S	L	M	M	S	S
CO 3	M	S	L	M	M	S	L	S	L	S
CO 4	L	S	S	L	S	M	S	L	S	M
CO 5	S	M	M	S	L	S	M	S	S	S

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Data Structures and Algorithms Practical		
Core–III - Practical	I Year & I Semester	Credit:3	436C1C

Objectives:

To provide the foundations of the practical implementation and usage of Algorithms and Data Structures.

To ensure that the student evolves into a competent programmer capable of designing and analysing implementations of algorithms and data structures for different kinds of problems.

To expose the student to the algorithm analysis techniques,

To make the students understand the theory of reductions, and to the classification of problems.

To make the students to be sure of complexity classes like NP.

Outcomes:

1.	Design and analyse programming problem statements.	K1, K2
2.	Choose appropriate data structures and algorithms, understand the ADT/libraries, and use it to design algorithms for a specific problem.	K1, K2
3.	Be familiar with programming language constructs available for rapid application development,	K3, K4
4.	Understand the necessary programmatic abstraction to solve problems.	K4, K5
5.	Gain the capacity to solve real life problems by matching to the available algorithms.	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

Implement the following using Java or Python

- Write a program to perform the following operations on a heterogeneous singly linked list. i) Creation ii) Insertion iii) Deletion iv) Traversal.
- Write a program to perform the following operations on a heterogeneous doubly linked list. i) Creation ii) Insertion iii) Deletion iv) Traversal in both ways
- Write a program that implements using java generic class, the stack (its operations)
- Write a program that implements using java generic class, the queue (its operations)
- Write a program that implements the Quick sort method.
- Write a program that implement the Merge sort method.
- Write a program that implement the SHELL sort method.
- Write a program to perform the following: i) Creating a Binary Tree of integers ii) Traversing the above binary tree in preorder, inorder and postorder.
- Write a program to perform the following: i) Creating a AVL Tree ii) insertion iii) deletion iv) Traversing the above AVL tree in preorder, inorder and postorder.
- Write a program that uses functions to perform the following: i) Creating a SplayTree ii) traverse

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

11. Write a program to perform the following: i) Creating a B-Tree of integers ii) insertion iii) deletion
12. Write a program that implements Kruskal's algorithm using a disjoint set data structure. The program takes as input a file (data.txt), in which each line either represents a vertex or an edge. For the edge lines, the first integer on that line representing the starting vertex, the second the ending vertex, and the third the weight of the edge. Use this file to construct, line by line, the graph upon which Kruskal's algorithm will be run (do NOT hardcode this graph!).
13. Write a program to simulate various graph traversing algorithms.
14. Write a program to find the minimal spanning tree of a graph using the Prim's algorithm.
15. Write a program to find shortest path using Bellman Ford's Algorithm

Recommended Texts:

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++:Pearson Education; 4th Edition,

Reference Books:

1. Sartaj Sahni, Data Structures Algorithms and Applications in C++, 2nd Edition, Universities Press, 2007.
2. Ellis Horowitz, Sartaj Sahni, Rajasekharan, Fundamentals of Algorithms, 2nd Edition, Universities Press, 2009.
3. Aho V Alfred, Hapcroft E John, Ullman D Jeffry, Data Structures and Algorithms, 1st Edition, Pearson Education, 2002.
4. Adam Drozdek, Thomson, Data Structures and Algorithms in JAVA, 3rd Edition, Cengage Learning, 2008.
5. Horowitz, Sahni, Mehta, Fundamentals of Data Structures in C++, 2nd Edition, Universities Press, 2007.

Web References:

1. <https://nptel.ac.in/courses/106102064>

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CO 1	S	M	M	S	M	S	L	S	L	M
CO 2	S	L	S	M	S	L	M	M	S	S
CO 3	M	S	L	M	M	S	L	S	L	S
CO 4	L	S	S	1	S	M	S	L	S	M
CO 5	S	M	M	S	L	S	M	S	L	S

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Python Programming		
Core–II - Theory	I Year & I Semester	Credit:4	436C1B

Objectives:

- To Provide advanced programming knowledge in python environment
- To Make interactive Python programs.
- To develop GUI based applications
- To utilise libraries and APIs for rapid application development
- To use python as an analytical tool for different mathematical models

Outcomes:

1.	Be able to program decorators, closures, lambda, iterators and generators comprehensions with in OOP.	K1, K2
2.	Learn modern data structures to include collections, array, and queues	K2, K3
3.	Use platform independent file manipulation, file pattern matching using CSV, HTML, XML, JASON	K3
4.	Be able to set up a client-server program and also multi-process applications.	K4, K5
5.	Be able to use python as an analytical and presentation tool	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I: Object Oriented Python – Class, Objects. member types (public. Private, protected) self, init method, method overloading. Inheritance, method overriding, Polymorphism, Encapsulation, Assertion, Decorators, Generators, Iterators.

UNIT II: Threading in Python: Thread, Starting a Thread, Daemon Threads, join() a Thread, Working With Many Threads, Using a Thread Pool Executor, Race Conditions, Basic Synchronization Using Lock, Deadlock, Producer-Consumer Threading, Producer Consumer Using Lock, Producer-Consumer Using Queue, Threading Objects, Semaphore, Timer

UNIT III: Database programming using Python: Connecting to a database (sqlite, mysql) using Python, Sending DML and DDL queries and processing the result from a Python Program. Network programming using Python: An introduction to client-server programming, Basics of TCP and UDP protocols, Introduction to socket programming, Building an HTTP client and server

UNIT IV: GUI in Python: Introduction to GUI building libraries, Widgets: Button - Canvas - Check button -Entry - Frame -Label – List box – Menu button - Menu - Message –Radio button - Scale - Scrollbar - Text – Top level – Spin box- Paned Window – Label Frame – Message Box

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Basic image processing using Python: Introduction to digital image processing, Basic operations on an image: Crop - Scale - Rotate - Flip - Changing contrast, brightness and color - Edge detection, blur, sharpening

UNIT V: Basic numerical processing using Python: Introduction to numpy , Creation of vectors and matrices, Matrix manipulation Basic data analysis using Python: Introduction to Pandas ,Pandas data structures – Series and DataFrame , Data wrangling using pandas : Loading a dataset into a dataframe- Selecting Columns from a dataframe - Selecting Rows from a dataframe - Adding new data in a dataframe - Deleting data from a dataframe Basic data visualization using : Introduction to Matplotlib, Scatter plot , Line plot, Bar chart, Histogram, Box plot.

Recommended Texts:

1. John Hunt; Advanced Guide to Python 3 Programming; Springer Nature Switzerland AG; 2019

Reference Books:

2. Eric Matthes, Python Crash Course: A Hands-On, Project-Based Introduction to Programming, 2nd Edition, No starch Press, 2019.
3. Mark Lutz; Learning Python, 5th Edition; O'Reilly Media, 2013
4. Mark Lutz, “Programming Python”, 4th edition, O'Reilly Media, 2010.

Web References:

1. <https://realpython.com/tutorials/advanced/>

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	M	S	S	S	L	M
CO 2	S	L	S	M	S	L	M	M	S	S
CO 3	M	S	L	M	M	S	L	S	L	S
CO 4	L	S	S	L	S	M	S	L	S	M
CO 5	S	M	M	S	L	S	M	S	S	S

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Python Programming Practical		
Core–IV - Practical	I Year & I Semester	Credit:3	436C1D

Objectives:

Design, implement and test readable, efficient programs that take advantage of Python built-in capabilities and follow Python best practices.

Understand implementation differences and performance tradeoffs associated with various Python data structures.

Develop Python applications using the modules and packages available in the Python standard library.

Develop Python applications using third party libraries.

Design, implement and test Python programs that include a graphical user interface, data analysis and visualization, web data extraction and web applications.

Outcomes:

1.	Apply exception handling and user defined exception(s) Develop Module(s) and Package(s) in python	K1, K2
2.	Possess an ability to write database applications in Python	K1, K2
3.	Implement Object Oriented concepts in programming Apply Collection modules for the data types	K3, K4
4.	Possess the Object-oriented programming skills in Python. and the skill of to design graphical-user interfaces (GUI) in Python.	K4, K5
5.	Make use of Pandas and Numpy Libraries	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

1) Exception

- Write a python program to catch following exception i) Value Error ii) Index Error iii) Name Error iv) Type Error v) DivideZero Error
- Write a python program to create user defined exceptions.
- Write a python program to understand the use of else and finally block with try block.
- Write a python program that uses raise and exception class to throw an exception.

2) Numpy Library

- Create a numpy array from list, tuple with float type
- Python program to demonstrate slicing, integer and boolean array indexing
- Write a python program to find min, max, sum, cumulative sum of array.
- Write a python program to demonstrate use of ndim, shape, size, dtype.

3) Numpy Library: Linear Algebra

- Write a python program to find rank, determinant, and trace of an array. b) Write a python program to find eigenvalues of matrices
- Write a python program to find matrix and vector products (dot, inner, outer, product), matrix exponentiation.

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

- d) Write a python program to solve a linear matrix equation, or system of linear scalar equations.
- e) Create a white image using NumPy in Python and
- f) Convert a NumPy array to an image and Convert images to NumPy array?
- g) Perform Sorting, Searching and Counting using Numpy methods.
- h) Write a program to demonstrate the use of the reshape() method.

4) Pandas Library

- a) Write a python program to implement Pandas Series with labels.
- b) Create a Pandas Series from a dictionary.
- c) Creating a Pandas DataFrame.
- d) Write a program which make use of following Pandas methods i) describe() ii) head() iii) tail()
- e) Write a program that converts Pandas DataFrame and Series into numpy.array.
- f) Write a program that demonstrates the column selection, column addition, and column deletion.
- g) Write a program that demonstrates the row selection, row addition, and row deletion.
- h) Get n-largest and n-smallest values from a particular column in Pandas dataframe

5) Visualization

- a) Write a program which use pandas inbuilt visualization to plot following graphs: i. Bar plots ii. Histograms iii. Line plots iv. Scatter plots
- b) Write a program to demonstrate use of groupby() method.
- c) Write a program to demonstrate pandas Merging, Joining and Concatenating
- d) Creating dataframes from csv and excel files.

6) Object Oriented Programming:

- a) Write a Python class named Person with attributes name, age, weight (kgs), height (ft) and takes them through the constructor and exposes a method get_bmi_result() which returns one of "underweight", "healthy", "obese"
- b) Write a python program to demonstrate various kinds of inheritance.
- c) Write a python program to demonstrate operator overloading.
- d) Write a python program to create abstract classes and abstract methods.

7) MULTITHREADING

- a. Write a python program to create two threads to keep a count of number of even numbers entered by the user.
- b. Write a JAVA program that creates threads by extending Thread class. First thread display "Good Morning "every 1 sec, the second thread displays "Hello "every 2 seconds and the third display "Welcome" every 3 seconds.

Recommended Texts:

1. Martin C. Brown (Author), "Python: The Complete Reference" McGraw Hill Education, Fourth edition , 2018

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Reference Books:

1. R. Nageswara Rao , “Core Python Programming” Dreamtech Press India Pvt Ltd 2018.

Web References:

1. <https://realpython.com/tutorials/advanced/>
2. https://onlinecourses.nptel.ac.in/noc19_cs40/preview
3. https://onlinecourses.nptel.ac.in/noc19_cs41/preview

Mapping with Programme Outcomes:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10
CO 1	S	M	M	S	M	L	S	S	L	M
CO 2	S	L	S	L	S	L	M	M	M	S
CO 3	M	S	L	M	M	S	L	S	L	S
CO 4	L	S	S	L	S	M	S	L	S	M
CO 5	S	M	M	S	L	S	M	S	S	S

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Computer Architecture		
Elective –I - Theory	I Year & I Semester	Credit:3	436E1C

Objectives:

To introduce the students to the recent trends in the field of Computer Architecture and identify performance related parameters.

To learn the different multiprocessor issues.

To expose the different types of multicore architectures.

To understand the design of the memory hierarchy.

Outcomes:

1.	Identify the limitations of Instruction-level parallelism (ILP).	K1, K2
2.	Discuss the issues related to multiprocessing and suggest solutions	K2, K3
3.	Point out the salient features of different multicore architectures and how they exploit parallelism.	K3, K4
4.	Discuss the various techniques used for optimizing the cache performance	K4, K5
5.	Design hierarchical memory system and Point out how data level parallelism is exploited in architectures	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I FUNDAMENTALS OF COMPUTER DESIGN AND ILP

Fundamentals of Computer Design – Measuring and Reporting Performance – Instruction Level Parallelism and its Exploitation – Concepts and Challenges –Exposing ILP - Advanced Branch Prediction - Dynamic Scheduling - Hardware-Based Speculation - Exploiting ILP - Instruction Delivery and Speculation - Limitations of ILP - Multithreading

UNIT II MEMORY HIERARCHY DESIGN

Introduction – Optimizations of Cache Performance – Memory Technology and Optimizations – Protection: Virtual Memory and Virtual Machines – Design of Memory Hierarchies – Case Studies.

UNIT III MULTIPROCESSOR ISSUES

Introduction- Centralized, Symmetric and Distributed Shared Memory Architectures – Cache Coherence Issues – Performance Issues – Synchronization – Models of Memory Consistency – Case Study-Interconnection Networks – Buses, Crossbar and Multi-stage Interconnection Networks

UNIT IV MULTICORE ARCHITECTURES

Homogeneous and Heterogeneous Multi-core Architectures – Intel Multicore Architectures – SUN CMP architecture – IBM Cell Architecture. Introduction to Warehouse-scale computers

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Architectures- Physical Infrastructure and Costs- Cloud Computing –Case Study- Google Warehouse-Scale Computer.

UNIT V VECTOR, SIMD AND GPU ARCHITECTURES

Introduction-Vector Architecture – SIMD Extensions for Multimedia – Graphics Processing Units – Case Studies – GPGPU Computing – Detecting and Enhancing Loop Level Parallelism-Case Studies.

Recommended Texts:

1. Darryl Gove ;Multicore Application Programming: For Windows, Linux, and Oracle Solaris, Pearson, 2011

Reference Books:

1. David B. Kirk, Wen-mei W. Hwu, – Programming Massively Parallel Processors, Morgan Kauffman, 2010
2. David E. Culler, Jaswinder Pal Singh, – Parallel computing architecture : A hardware/software approach , Morgan Kaufmann /Elsevier Publishers, 1999
3. John L. Hennessey and David A. Patterson, – Computer Architecture – A Quantitative Approach, Morgan Kaufmann / Elsevier, 5th edition, 2012.
4. Kai Hwang and Zhi.Wei Xu, – Scalable Parallel Computing, Tata McGraw Hill, NewDelhi, 2003

Web References:

1. <https://www.youtube.com/watch?v=v7iefsovo9M&list=PLwdnzlV3ogoWJhBxBY u-K4l-q-nNHd24D>

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	M	S	S	M	L	M	L	S
CO2	S	M	S	L	M	L	M	L	M	S
CO3	S	S	L	M	S	S	L	M	L	M
CO4	M	L	S	M	M	L	S	L	M	S
CO5	S	S	M	S	L	M	L	M	S	L

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Cloud Computing		
Elective –I - Theory	I Year & I Semester	Credit:3	436E1A

Objectives:

- Gain knowledge on cloud computing, cloud services, architectures and applications.
- Enable the students to learn the basics of cloud computing with real time usage
- How to store and share, in and from cloud?

Outcomes:

1.	Understand the concepts of Cloud and its services	K1,K2
2.	Collaborate Cloud for Event & Project Management	K3,K4
3.	Analyze on cloud in Word Processing, Spread Sheets, Mail, Calendar, Database	K4,K5
4.	Analyze cloud in social networks	K5,K6
5.	Explore cloud storage and sharing	K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I:INTRODUCTION Cloud Computing Introduction, From, Collaboration to cloud, Working of cloud computing, pros and cons, benefits, developing cloud computing services, Cloud service development, discovering cloud services.

UNIT II:CLOUD COMPUTING FOR EVERYONE Centralizing email communications, cloudcomputing for community, collaborating on schedules, collaborating on group projects and events, cloud computing for corporation, mapping, schedules, managing projects, presenting on road.

UNIT III:USING CLOUD SERVICES Collaborating on calendars, Schedules and task management, exploring on line scheduling and planning, collaborating on event management, collaborating on contact management, collaborating on project management, collaborating on word processing, spreadsheets, and databases.

UNIT IV:OUTSIDETHECLOUD Evaluating webmail services, evaluating instant messaging, Evaluating web conference tools, creating groups on social networks, Evaluating onlinegroupware, collaborating via blog sand wikis.

UNIT V:STORING AND SHARING Understanding cloud storage, evaluating on line file storage, exploring on-line book-marking services, exploring on line photo editing applications, exploring photo sharing communities, controlling it with web based desktops.

Text Books

Michael Miller, “Cloud Computing”, Pearson Education, New Delhi, 2009.

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

ReferenceBooks

Anthony T. Velte, “Cloud Computing: A Practical Approach”, 1st Edition, Tata McGraw Hill Education Private Limited, 2009.

Related Online Contents

[MOOC,SWAYAM,NPTEL,Websitesetc.]

<https://nptel.ac.in/courses/106/105/106105167/>

https://www.tutorialspoint.com/cloud_computing/index.htm

<https://www.javatpoint.com/cloud-computing-tutorial>

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	L	S	M	S	M	S	M	M	M	S
CO2	M	S	M	S	S	S	M	M	M	S
CO3	S	S	S	S	S	S	S	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	M	S	S	S	S	S	S	S	S	S

*S-Strong; M-Medium; L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Internet of Things		
Elective –I - Theory	I Year & I Semester	Credit:3	436E1B

Objectives:

- About Internet of Things where various communicating entities are controlled and managed for decision making in the application domain.
- Enable students to learn the Architecture of IoT and IoT Technologies
- Developing IoT applications and Security in IoT, Basic Electronics for IoT, Arduino IDE, Sensors and Actuators Programming NodeMCU using Arduino IDE.

Outcomes:

1.	Understand about IoT, its Architecture and its Applications	K1,K2
2.	Understand basic electronics used in IoT & its role	K2,K3
3.	Develop applications with using Arduino IDE	K4
4.	Analyze about sensors and actuators	K5,K6
5.	Design IoT in real time applications using today's internet & wireless technologies	K6
K1-Remember; K2-Understand; K3-Apply; K4-Analyze; K5-Evaluate; K6-Create		

UNIT I: Introduction to IoT: Evolution of IoT – Definition & Characteristics of IoT - Architecture of IoT – Technologies for IoT – Developing IoT Applications – Applications of IoT – Industrial IoT – Security in IoT

UNIT II: Basic Electronics for IoT: Electric Charge, Resistance, Current and Voltage – Binary Calculations – Logic Chips – Microcontrollers – Multipurpose Computers – Electronic Signals – A/D and D/A Conversion – Pulse Width Modulation.

UNIT III: Programming Fundamentals with C using Arduino IDE: Installing and Setting up the Arduino IDE – Basic Syntax – Data Types/ Variables/ Constant – Operators – Conditional Statements and Loops – Using Arduino C Library Functions for Serial, delay and other invoking Functions – Strings and Mathematics Library Functions.

UNIT IV: Sensors and Actuators: Analog and Digital Sensors – Interfacing temperature sensor, ultrasound sensor and infrared (IR) sensor with Arduino – Interfacing LED and Buzzer with Arduino.

UNIT V: Sending Sensor Data Over Internet: Introduction to ESP8266 NODEMCU WiFi Module – Programming NODEMCU using Arduino IDE – Using WiFi and NODEMCU to transmit data from temperature sensor to Open Source IoT cloud platform (ThingSpeak).

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Text Books

1. Arshdeep Bahga, Vijay Madisetti, “Internet of Things: A Hands-On Approach”, 2014. ISBN: 978-0996025515
2. Boris Adryan, Dominik Obermaier, Paul Fremantle, “The Technical Foundations of IoT”, Artech Houser Publishers, 2017

Reference Books

1. Michael Margolis, “Arduino Cookbook”, O Reilly, 2011
2. Marco Schwartz, “Internet of Things with ESP8266”, Packt Publishing, 2016.
3. Dhivya Bala, “ESP8266: Step by Step Tutorial for ESP8266 IoT, Arduino NODEMCU Dev. Kit”, 2018.

Related Online Contents

1. https://onlinecourses.nptel.ac.in/noc20_cs66/preview
2. <https://www.javatpoint.com/iot-internet-of-things>
3. https://www.tutorialspoint.com/internet_of_things/index.htm

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	M	M	S	M	S	M	M	S	M
CO2	M	S	M	S	M	S	M	S	S	S
CO3	S	S	S	S	M	S	M	S	S	S
CO4	S	S	S	S	S	S	S	S	S	S
CO5	S	S	S	S	S	S	S	S	S	S

*S-Strong; M-Medium; L-Low

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Principles of Compiler Design		
Elective – II - Theory	I Year & I Semester	Credit:3	436E1D

Objectives:

- To learn the various phases of compiler.
- To learn the various parsing techniques.
- To understand intermediate code generation and run-time environment.
- To learn to implement the front-end of the compiler.
- To learn to implement code generators.

Outcomes:

1.	Understand the different phases of the compiler.	K1, K2
2.	Design a lexical analyzer for a sample language.	K2, K3
3.	Apply different parsing algorithms to develop the parsers for a given grammar.	K3, K4
4.	Design and implement a scanner and a parser using LEX and YACC tools	K4, K5
5.	Learn to implement code optimization techniques and a simple code generator.	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I: INTRODUCTION TO COMPILERS: Structure of a compiler – Lexical Analysis – Role of Lexical Analyzer – Input Buffering – Specification of Tokens – Recognition of Tokens – Lex – Finite Automata – Regular Expressions to Automata – Minimizing DFA.

UNIT II: SYNTAX ANALYSIS: Role of Parser – Grammars – Error Handling – Context free grammars – Writing a grammar – Top Down Parsing - General Strategies Recursive Descent Parser Predictive Parser-LL(1) Parser-Shift Reduce Parser-LR Parser- LR (0)Item Construction of SLR Parsing Table - Introduction to LALR Parser - Error Handling and Recovery in Syntax Analyzer-YACC.

UNIT III: INTERMEDIATE CODE GENERATION: Syntax Directed Definitions, Evaluation Orders for Syntax Directed Definitions, Intermediate Languages: Syntax Tree, Three Address Code, Types and Declarations, Translation of Expressions, Type Checking.

UNIT IV: RUN-TIME ENVIRONMENT AND CODE GENERATION: Storage Organization, Stack Allocation Space, Access to Non-local Data on the Stack, Heap Management - Issues in Code Generation - Design of a simple Code Generator.

UNIT V: CODE OPTIMIZATION: Principal Sources of Optimization – Peep-hole optimization - DAG- Optimization of Basic Blocks- Global Data Flow Analysis - Efficient Data Flow Algorithm.

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Recommended Texts:

1. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Compilers: Principles, Techniques and Tools; Pearson Education; Second Edition; 2013

Reference Books:

1. Randy Allen, Ken Kennedy, Optimizing Compilers for Modern Architectures: A Dependence based Approach, Morgan Kaufmann Publishers, 2002.
2. Steven S. Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufmann Publishers - Elsevier Science, India, Indian Reprint 2003.
3. Keith D Cooper and Linda Torczon, Engineering a Compiler, Morgan Kaufmann Publishers Elsevier Science, 2004.
4. V. Raghavan, Principles of Compiler Design, Tata McGraw Hill Education Publishers, 2010.
5. Allen I. Holub, Compiler Design in C, Prentice-Hall Software Series, 1993.

Web References:

1. https://www.youtube.com/watch?v=k4QXWFZZq1E&list=PLENQMW_c1dimxH Uu6KjuBC2rOlAaoLozF

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	S	L	S	M	L	M	L	L
CO2	S	M	S	S	S	M	M	L	M	L
CO3	S	S	L	M	S	S	L	M	L	M
CO4	M	L	S	M	M	L	S	L	M	S
CO5	S	S	M	S	L	M	M	L	M	L

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Natural Language Processing		
Elective – II - Theory	I Year & I Semester	Credit:3	436E1E

Objectives:

- To learn the fundamentals of natural language processing
- To understand the use of CFG and PCFG in NLP
- To understand the role of semantics of sentences and pragmatics
- To apply the NLP techniques to IR applications

Outcomes:

1.	To tag a given text with basic Language features	K1
2.	To design an innovative application using NLP components	K2
3.	To implement a rule-based system to tackle morphology/syntax of a language	K3, K4
4.	To design a tag set to be used for statistical processing for real-time applications	K4, K5
5.	To compare and contrast the use of different statistical approaches for different types of NLP applications.	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I INTRODUCTION: Origins and challenges of NLP – Language Modeling: Grammar-based LM, Statistical LM - Regular Expressions, Finite-State Automata – English Morphology, Transducers for lexicon and rules, Tokenization, Detecting and Correcting Spelling Errors, Minimum Edit Distance

UNIT II WORD LEVEL ANALYSIS: Unsmoothed N-grams, Evaluating N-grams, Smoothing, Interpolation and Backoff – Word Classes, Part-of-Speech Tagging, Rule based, Stochastic and Transformation-based tagging, Issues in PoS tagging – Hidden Markov and Maximum Entropy models.

UNIT III SYNTACTIC ANALYSIS: Context-Free Grammars, Grammar rules for English, Treebanks, Normal Forms for grammar – Dependency Grammar – Syntactic Parsing, Ambiguity, Dynamic Programming parsing – Shallow parsing – Probabilistic CFG, Probabilistic CYK, Probabilistic Lexicalized CFGs - Feature structures, Unification of feature structures.

UNIT IV SEMANTICS AND PRAGMATICS: Requirements for representation, First Order Logic, Description Logics – Syntax-Driven Semantic analysis, Semantic attachments – Word Senses, Relations between Senses, Thematic Roles, selection restrictions – Word Sense Disambiguation, WSD using Supervised, Dictionary & Thesaurus, Bootstrapping methods – Word Similarity using Thesaurus and Distributional methods.

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

UNIT V DISCOURSE ANALYSIS AND LEXICAL RESOURCES: Discourse segmentation, Coherence – Reference Phenomena, Anaphora Resolution using Hobbs and Centering Algorithm – Coreference Resolution – Resources: Porter Stemmer, Lemmatizer, Penn Treebank, Brill's Tagger, WordNet, PropBank, FrameNet, Brown Corpus, British National Corpus (BNC).

Recommended Texts:

1. Daniel Jurafsky, James H. Martin; Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech; Pearson Publication; 2014.
2. Steven Bird, Ewan Klein and Edward Loper, – Natural Language Processing with Python , First Edition, O'Reilly Media, 2009.

Reference Books:

1. Breck Baldwin, – Language Processing with Java and LingPipe Cookbook, Atlantic Publisher, 2015.
2. Richard M Reese, – Natural Language Processing with Java ,O Reilly Media, 2015.
3. Nitin Indurkha and Fred J. Damerau, – Handbook of Natural Language Processing, Second Edition, Chapman and Hall/CRC Press, 2010.
4. Tanveer Siddiqui, U.S. Tiwary, – Natural Language Processing and Information Retrieval, Oxford University Press, 2008.

Web References:

1. https://www.youtube.com/watch?v=oWsMIW5xUc&list=PLLssT5z_DsK8HbD2sPcUIDfQ7zmBarMYv

Mapping with Programme Outcomes:

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CO1	M	S	S	L	S	M	L	M	L	L
CO2	S	M	M	L	M	L	M	L	M	S
CO3	S	S	L	M	S	S	L	M	L	M
CO4	M	L	S	M	M	L	S	L	M	S
CO5	S	S	M	S	L	M	L	M	M	L

S-Strong M-Medium L-Low

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Distributed Database Systems		
Elective – II - Theory	I Year & I Semester	Credit:3	436E1F

Objectives:

To introduction students to Distributed DBMS and associated problems.

To make students understand various algorithms and techniques for managing distributed database.

To understand theoretical and practical aspects of distributed database systems.

To study and identify various issues related to the development of distributed database system.

To make students understand Transaction Management & Compare various approaches to concurrency control in Distributed database

Outcomes:

1.	Apply various fragmentation techniques given a problem	K1
2.	Analyse and calculate the cost of enforcing semantic integrity control	K2, K3
3.	Use the steps of query processing	K4
4.	Apply optimization techniques are applies to Distributed Database	K4, K5
5.	Apply effectively Query Optimization Algorithms	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

Unit-I: Introduction: Distributed Data Processing, Distributed Database Systems, Promises of DDBSs, Complicating factors, Problem areas Overview of RDBMS: Concepts, Integrity, Normalization - Distributed DBMS Architecture: Autonomy, Distribution, Heterogeneity DDBMS Architecture – Client/Server, Peer to peer, MDBS

Unit-II: Data Distribution Alternatives: Design Alternatives – localized data, distributed data Fragmentation – Vertical, Horizontal (primary & derived), hybrid, general guidelines, correctness rules Distribution transparency – location, fragmentation, replication Impact of distribution on user queries – No Global Data Dictionary (GDD), GDD containing location information Example on fragmentation

Unit-III: Semantic Data Control: View Management, Authentication – database authentication, OS authentication, Access Rights, Semantic Integrity Control – Centralized & Distributed, Cost of enforcing semantic integrity - : Query Processing: Query Processing Problem, Layers of Query Processing Query Processing in Centralized Systems – Parsing & Translation, Optimization, Code generation, Example Query Processing in Distributed Systems – Mapping global query to local, Optimization,

UNIVERSITY OF MADRAS
M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE
SYLLABUS WITH EFFECT FROM 2023-2024

Unit-IV: Optimization of Distributed Queries: Query Optimization, Centralized Query Optimization, Join Ordering Distributed Query Optimization Algorithms - Distributed Transaction Management & Concurrency Control: Transaction concept, ACID property, Objectives of transaction management, Types of transactions, Objectives of Distributed Concurrency Control, Concurrency Control anomalies, Methods of concurrency control, Serializability and recoverability, Distributed Serializability, Enhanced lock based and timestamp based protocols, Multiple granularity, Multi version schemes, Optimistic Concurrency Control techniques

Unit-V: Distributed Deadlock & Recovery: Deadlock concept, Deadlock in Centralized systems, Deadlock in Distributed Systems – Detection, Prevention, Avoidance, Wait-Die Algorithm, Wound-Wait algorithm Recovery in DBMS - Types of Failure, Methods to control failure, Different techniques of recoverability, Write- Ahead logging Protocol, Advanced recovery techniques- Shadow Paging, Fuzzy checkpoint, ARIES, RAID levels, Two Phase and Three Phase commit protocols

Recommended Texts:

1. Ozsu; Principles of Distributed Database Systems; Springer; 4th edition;2020

Reference Books:

1. Rahimi &Haug; Distributed Database Management Systems;Wiley;2010
2. Distributed Database Systems, Chanda Ray, Pearson Publication
3. Sachin Deshpande; Distributed Databases; Dreamtech;2014

Web References:

1. <https://www.youtube.com/watch?v=dIBVWMdGhqw&list=PLUJ7JmcrTifBROWODSG8wgyl20XgBuE-N>

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	M	S	S	M	L	M	L	S
CO2	S	M	S	L	M	L	M	L	M	S
CO3	S	S	L	M	S	S	L	M	L	M
CO4	M	L	S	M	M	L	S	L	M	S
CO5	S	S	M	S	L	M	L	M	S	L

S-Strong M-Medium L-Low